

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

1st Semester of B. Pharm. Examination

University Theory Examination January 2018

PH116 Pharmaceutical Biology-I

Date: 05.01.2018, Friday

Time: 10:00 a.m. to 1:00 p.m.

Maximum Marks: 80

Instructions:

1. There are three sections in this question paper.
2. SECTION – I comprises of Question 1. Total marks for Section 1 are 20. There are 20 sub-questions (MCQ type). Answers to SECTION – I are to be given in Answer Sheet for MCQ type questions provided to you. Maximum time allotted for SECTION – I is 30 minutes. Answers to SECTION – I must be written during the first 30 minutes of the examination.
3. Answers to SECTION – II and SECTION – III are to be provided in separate Main Answer Books provided to you.
4. Figures to right indicate marks.
5. Draw neat labeled sketches wherever necessary.

SECTION - I

Q 1 Attempt all questions. Each question is of one mark.

20

1. All of the following substances move in & out of cell except:
[A] Nutrients
[B] Gases
[C] Waste
[D] Blood
2. The nucleus contains DNA molecules arranged in bundles called:
[A] Proteins
[B] Gametes
[C] Cytoplasm
[D] Chromosomes
3. When the concentration of a substance is higher on one side of the cell's permeable barrier, its molecules use osmosis or diffusion to move through the barrier without the cell using any energy. This process is called:
[A] Passive transport
[B] Active transport
[C] Mitosis
[D] Replication
4. When cells divide and multiply in the embryo and change in shape and structure, the process is called:
[A] Duplication
[B] Differentiation
[C] Replication
[D] Osmosis
5. A microbe is introduced into an environment where its natural metabolism results in the detoxification or break down of hazardous chemicals or pollutants is called as

- [A] Biodistribution
- [B] Bioremediation
- [C] Bio-fixation
- [D] Bio-hazard

6. Following part of the eye regulates the amount of light entering the eyeball through the pupil:
- [A] Cornea
 - [B] Iris
 - [C] Retina
 - [D] Choroid
7. The following is not an acidic dye
- [A] Malachite Green
 - [B] Eosin
 - [C] Acid Fuschin
 - [D] Niosin
8. Taste Buds are found in the _____.
- [A] Olfactory epithelium
 - [B] Papillae
 - [C] Gustatory epithelium
 - [D] Epiglottis
9. Endospore of the microorganism can be identified by
- [A] Simple staining
 - [B] Gram +Ve staining
 - [C] Gram -Ve staining
 - [D] Special staining technique
10. These voluntary muscles are not striated.
- [A] Skeletal
 - [B] Cardiac
 - [C] Smooth
 - [D] Facial
11. How many mitochondrions are present in liver?
- [A] >2000
 - [B] 200-500
 - [C] 1000-1500
 - [D] 500-800
12. The study of morphology of human body part is called as
- [A] Physiology
 - [B] Cell activity
 - [C] Cell functions
 - [D] Anatomy
13. The action of alcohol during Gram staining is
- [A] Decolorizes the cells
 - [B] Allows the color to stain
 - [C] Stain the cell
 - [D] It adds color

14. Night blindness can be treated with
 [A] Vitamin C
 [B] Vitamin D
 [C] Vitamin A
 [D] Vitamin K
15. Which are the 3 types photopigments of cones?
 [A] Red, green and blue
 [B] Red, green and yellow
 [C] Red, orange and blue
 [D] Red, yellow and blue
16. Nervous tissue consists of _____
 [A] Neurons
 [B] Neurons and Neuroglia
 [C] Neuroglia
 [D] Connective tissue
17. Why can cardiac muscle fibers contract for longer periods than skeletal muscle fibers?
 [A] Extracellular calcium partially controls the strength (and length) of contraction
 [B] Cardiac muscle is uninucleate rather than multinucleate
 [C] T-tubules is more developed in cardiac muscle
 [D] Cardiac muscle is self-excitatory
18. The phycologist is the person who is involved in the study of
 [A] Algae
 [B] Fungi
 [C] Virus
 [D] Bacteria
19. In bacterial cells which one is used for movement?
 [A] Nucleoid
 [B] Cell Wall
 [C] Flagellum
 [D] Capsule
20. Following is a contracting protein.
 [A] Titin
 [B] Troponin
 [C] Tropomyosin
 [D] Myosin

SECTION – II

- Q 2** Attempt any **TWO** of the following
- | | | |
|----------|---|----|
| A | Enlist the various staining techniques for identification of microbes and explain 'simple staining' in brief. | 05 |
| B | Define: tissue. Write a brief note on 'Nervous tissue'. | 05 |
| C | Explain the following:
(i) Isolation of micro-organism (ii) Incubation of micro-organism | 05 |

- Q 3** Attempt any **TWO** of the following
- A** Recognize the role of 'Gram staining technique' in identification of micro-organism and explain in brief. 05
 - B** Differentiate: Epithelial tissue & muscular tissue. 05
 - C** Draw neat and clean labelled diagram of ear indicating the physiology of hearing. 05

SECTION – III

- Q 4** Attempt any **FOUR** of the following
- A** Describe the various functions of Cell Membrane. 05
 - B** Explain the various properties of skeletal muscles. 05
 - C** Draw neat and clean labelled diagram of 'Cross-section of Eye'. 05
 - D** How the skin helps in 'Thermoregulation' of body? 05
 - E** Define cell. Describe the various parts of cell with labeled diagram. 05
 - F** Write a note on: Eukaryotic cells. 05
- Q 5** Attempt any **FOUR** of the following
- A** What do you mean by micro-organism? What are the advantages and disadvantages of it? 05
 - B** Discuss the physiology of muscle contraction in detail. 05
 - C** Enumerate the different transport mechanism across cell membrane and explain any one in detail. 05
 - D** Classify various connective tissues and explain in brief. 05
 - E** Write a detailed note on cell cycle. 05
 - F** Explain the different functions of skin. 05
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